

HAUSMAN AND BARTIK INSTRUMENTS

Empirical Proxies with Equilibrium Objects

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May 2026

We want to instrument an endogenous regressor with an **exogenous shock**.

- **Hausman (1996)**: instrument price p_{mt} with average price in **other** markets, $h_{mt} = \frac{1}{M-1} \sum_{m' \neq m} p_{m't}$
- **Bartik (1991)**: instrument labor market growth ΔL_ℓ with $B_\ell = \sum_n s_{\ell n} g_n$, predicted growth using national shares

Neither is an external natural experiment. Ideally, both are **empirical proxies** for unobserved shocks (η):

- prices reflect (unobserved) cost shocks
- national sector growth reflects (unobserved) sector demand shocks

The instrument is constructed from **equilibrium outcomes**.

Common form: $z_i = w_i'g$, with $g = h(\varepsilon, \eta)$ an equilibrium outcome.

Hausman

- Endogenous: p_m
- Instrument: $z_m = \frac{1}{M-1} \sum_{m' \neq m} p_{m'}$
- Proxy stands in for: cost shock η
- Weights w_i : $\frac{1}{M-1}$ **deterministic**
- g : observed prices across markets

Bartik / shift-share

- Endogenous: ΔL_ℓ
- Instrument: $z_\ell = \sum_n s_{\ell n} g_n$
- Proxy stands in for: sector demand η_n
- Weights w_i : $s_{\ell n}$ **stochastic**
- g : observed national growth rates

Usual argument is **statistical**:

- if shocks ε and η are independent across markets/regions, the proxy is exogenous (see e.g. Hausman Bresnahan debate)

Our point: this reasoning is incomplete. The **economic structure** of the proxy matters — captured by the Jacobian

$$J_{\varepsilon} \equiv \frac{\partial h}{\partial \varepsilon'}$$

i.e.

the reduced-form response of the proxy to the structural shock ε .

And when weights are stochastic, $\text{Cov}(w_i, \varepsilon_i)$ matters too.

1. Frame Hausman and shift-share as two instances of $z_i = w_i'g$ with $g = h(\varepsilon, \eta)$
2. Unified bias formula

$$\mathbb{E} \left[\sum_i z_i \varepsilon_i \right] = \underbrace{\bar{g}' c_{w\varepsilon}}_{\text{weight bias}} + \underbrace{\text{tr}(\mathbb{E}[W J_\varepsilon] \Sigma_\varepsilon)}_{\text{contamination}}$$

3. Hausman = JIVE: only the **contamination** channel is active
4. Bartik: both channels can be active
5. Diagnostics, and apply to DellaVigna and Gentzkow (2019)

1. Running examples: monopoly across markets; regions across sectors
2. Setup: $z_i = w_i'g$, $g = h(\varepsilon, \eta)$
3. Unified bias formula
4. Hausman as the deterministic- W case
5. Bartik as the stochastic- W case (BHJ + AKM)
6. Four channels, four corrections
7. Diagnostics + illustration

Running examples

- Three markets (Chicago, Milwaukee, Detroit), T periods
- Demand: $q_{mt} = \alpha + \beta p_{mt} + \xi_{mt}$, with ξ_{mt} iid, $\text{Var}(\xi) = \sigma_\xi^2$

Multi-market monopolist with a price-dispersion penalty.

Demand $q_m = \xi_m - \alpha p_m$, MC $c_m = \alpha_m^c + \eta_t$.

$$\max_{\{p_m\}} \sum_m \underbrace{(p_m - c_m)(\xi_m - \alpha p_m)}_{\text{per-market profit}} - \frac{\kappa}{2} \sum_m (p_m - \bar{p})^2$$

κ captures any tendency toward uniform pricing: menu costs, fairness norms, managerial simplicity (DellaVigna–Gentzkow 2019).

FOC at each m : $(2\alpha + \kappa)p_m = \xi_m + \alpha c_m + \kappa \bar{p}$.

Solving the fixed point:

$$p_{mt} = \underbrace{\alpha_m^c/2}_{\alpha_m} + \underbrace{\frac{\alpha + \kappa/2}{2\alpha + \kappa}\eta_t}_{\eta_t} + \underbrace{\frac{1}{2\alpha + \kappa}}_{\delta} \xi_{mt} + \underbrace{\frac{\kappa}{2\alpha(2\alpha + \kappa)}}_{\lambda} \bar{\xi}_t$$

- $\kappa = 0$: independent monopoly $\rightarrow \lambda = 0$, no Hausman bias
- $\kappa > 0$: uniform-pricing tendency $\rightarrow \lambda > 0 \rightarrow$ cross-market J_ξ entries

- Reduced-form pricing with uniform pricing:

$$p_{mt} = \alpha_m + \eta_t + \delta \xi_{mt} + \lambda \bar{\xi}_t, \quad \bar{\xi}_t = \frac{1}{M} \sum_m \xi_{mt}$$

- η_t : cost shock (the shock we **want** to use)
- $\delta \xi_{mt}$: own-market endogeneity
- $\lambda \bar{\xi}_t$: cross-market spillover (uniform-pricing tendency)

Hausman LOO: $h_{mt} = \frac{1}{M-1} \sum_{m' \neq m} p_{m't}$ — proxy for η_t .

Even with ξ independent across markets, leaving out the focal market does **not** purge h_{mt} of ξ_{C1} :

$$\text{Cov}(\xi_{C1}, h_{C1}) = \text{Cov}(\xi_{C1}, \lambda \bar{\xi}_1) = \frac{\lambda \sigma_{\xi}^2}{3}$$

- The statistical assumption (independence of ξ) is fine.
- The **economic** problem: $\partial \frac{p_{M1}}{\partial} \xi_{C1} = \frac{\lambda}{M} \neq 0$.

Cross-market spillovers, not own-market contamination.

- Structural equation: $\hat{\omega}_\ell = \tilde{\varphi} \hat{L}_\ell + \varepsilon_\ell$, L regions ℓ , N sectors n
- From BHJ model, region-sector employment is

$$\hat{L}_{\ell n} = g_n^* + \frac{\sigma}{\sigma + \varphi} \varepsilon_\ell + \hat{\xi}_{\ell n} - \frac{\sigma}{\sigma + \varphi} \sum_{n'} s_{\ell n'} (g_{n'}^* + \hat{\xi}_{\ell n'})$$

- Aggregated: $\hat{g}_n = (L - 1)^{-1} \sum_{l' \neq l} \hat{L}_{\ell n}$ – **equilibrium proxy**, not a primitive.
- Instrument: $z_\ell = \sum_n s_{\ell n} \hat{g}_n$
 - $s_{\ell n}$: lagged share of sector n in region ℓ
- Key **modeling** assumptions: immobile labor, no trade, and supply residual

Setup

L units indexed by i . Structural eq. $y_i = \beta x_i + \varepsilon_i$.

Instrument.

$$z_i = w_i' g, \quad g = (g_1, \dots, g_K)' \in \mathbb{R}^K, \quad w_i \in \mathbb{R}^K$$

Stack rows: W is $L \times K$.

- Hausman: $K = MT$, g is the price vector
- Bartik: $K = N$, g is the vector of national sector growth rates
 - Can relax to panel

Assumption. The proxy is an equilibrium outcome of two primitive shocks:

$$g = h(\varepsilon, \eta), \quad \mathbb{E}[\varepsilon] = 0, \quad \eta \perp \varepsilon$$

- $\varepsilon = (\varepsilon_1, \dots, \varepsilon_L)$: structural residual (mean-zero)
- $\eta = (\eta_1, \dots, \eta_R)$: exogenous shifters (mean **not** assumed zero)

Two Jacobians (at $\varepsilon = 0, \eta - \mathbb{E}[\eta] = 0$):

$$J_\varepsilon \equiv \frac{\partial h}{\partial \varepsilon'} \quad (K \times L), \quad J_\eta \equiv \frac{\partial h}{\partial \eta'} \quad (K \times R)$$

- J_η : **relevance**
- J_ε : **contamination** of the proxy by the residual

The deterministic baseline:

$$\bar{g} \equiv \mathbb{E}[g] = h(0, \mathbb{E}[\eta]) + O(\text{Var}(\varepsilon, \eta))$$

$\bar{g}$ combines the no-shock equilibrium **and** the contribution of $\mathbb{E}[\eta]$.

Hausman. $g = p$.

- $\bar{g} = \bar{p}_m = \alpha_m + \mathbb{E}[\eta_t]$
- Nonzero, varies across markets

Bartik. g is sector growth.

- $h(0, 0) = 0$ (“no shocks, no growth”)
- But $\bar{g}_n = (J_\eta \mathbb{E}[\eta])_n$ may vary across sectors (tech vs. steel)

Key assumption in BHJ (2022): $E(g_n \mid \bar{\varepsilon}, s) = \mu$ for all n .

Assumption. W is either

- **deterministic** — fixed function of the design (Hausman: panel index $\rightarrow$ JIVE)
- **stochastic** — with $\eta \perp (\varepsilon, W)$ (Bartik: lagged exposure shares)

This distinction distinguishes the two designs:

- Hausman (W deterministic): $\sum_i w_i \mathbb{E}[\varepsilon_i] = 0$ mechanically
- Bartik (W stochastic): $\text{Cov}(w_i, \varepsilon_i)$ can be nonzero

The bias formula

Proposition. To first order in (ε, η) :

$$\mathbb{E} \left[\sum_i z_i \varepsilon_i \right] = \underbrace{\bar{g}' c_{w\varepsilon}}_{\text{(I) share endogeneity}} + \underbrace{\text{tr}(\mathbb{E}[W J_\varepsilon] \Sigma_\varepsilon)}_{\text{(II) contamination}} + R$$

where $c_{w\varepsilon} \equiv \sum_i \mathbb{E}[w_i \varepsilon_i]$.

Proof sketch. Write $\eta = \mathbb{E}[\eta] + \eta^*$, $\eta^* \perp (\varepsilon, W)$, expand

$$g \approx \bar{g} + J_\varepsilon \varepsilon + J_\eta \eta^*$$

and take expectations. Term in η^* vanishes by independence + mean zero.

Hausman. W deterministic $\rightarrow c_{w\varepsilon} = \sum_i w_i \mathbb{E}[\varepsilon_i] = 0$.

$$\mathbb{E}[\xi' W p] = \text{tr}(W J_\varepsilon) \Sigma_\varepsilon$$

Only contamination. $\bar{g}$ **irrelevant** for the IV moment.

Bartik, predetermined shares ($W \perp \varepsilon$): $c_{w\varepsilon} = 0$. Contamination only.

Bartik, endogenous shares ($W \not\perp \varepsilon$): both terms active.

$$\bar{g}' c_{w\varepsilon} = \sum_n \bar{g}_n \underbrace{\sum_\ell \text{Cov}(s_{\ell n}, \varepsilon_\ell)}_{\text{share endogeneity}}$$

Bias scales with sector-specific mean growth $\bar{g}_n$.

Corollary. If shares sum to one, then term I vanishes whenever $\bar{g}_n$ is **constant across** n .

- Constants are absorbed by the regression intercept.
- Variation in $\bar{g}_n$ — high-growth tech vs. low-growth steel — is what produces bias.
- This is precisely what **recentering** (Borusyak–Hull) targets.

In Hausman, $\bar{g}_m$ also varies (across markets) — but term (I) is zero by construction.

Hausman: deterministic- W case

Per-period Jacobian for $p_t = g(\xi_t, \eta_t)$:

$$\left[J_t^\xi \right]_{m'm} = \delta \cdot \mathbb{1}\{m = m'\} + \frac{\lambda}{M}$$

- Diagonal δ : standard own-market endogeneity
- Off-diagonal $\frac{\lambda}{M}$: cross-market spillover via $\bar{\xi}_t$

Three canonical cases:

- **Independent pricing:** J_t^ξ diagonal
- **Uniform pricing:** all rows equal — maximal spillovers
- **Partial spillovers:** $\left[J_t^\xi \right]_{m'm} = \frac{\lambda}{M}$ off-diagonal

Linear IV: $\hat{\beta} = (p' C p)^{-1} (p' C q)$.

- 2SLS: $C = P_Z = Z(Z' Z)^{-1} Z'$
- JIVE: $C_{\text{JIVE}} = (I - \text{diag}(P_Z))^{-1} (P_Z - \text{diag}(P_Z))$
- CJIVE: replaces diag with within-cluster block

With Z = time dummies, the JIVE predicted price is exactly the LOO mean:

$$\hat{p}_{mt}^{\text{JIVE}} = \frac{1}{M-1} \sum_{m' \neq m} p_{m't} = h_{mt}$$

Hausman IV is JIVE. (Hahn et al. (2025))

$$\mathbb{E}[\xi' C_{\text{JIVE}} p] = \frac{\sigma_{\xi}^2}{M-1} \sum_t \underbrace{\left(\mathbf{1}' J_t^{\xi} \mathbf{1} - \text{tr}(J_t^{\xi}) \right)}_{\text{off-diagonal spillovers}}$$

For the example: $\mathbf{1}' J_t^{\xi} \mathbf{1} - \text{tr}(J_t^{\xi}) = M\lambda$, so per-period bias is

$$\frac{M}{M-1} \lambda \sigma_{\xi}^2 \rightarrow \lambda \sigma_{\xi}^2 \quad \text{as } M \rightarrow \infty$$

- $\lambda = 0$ (independent pricing): no bias
- $\bar{g} = \bar{p}_m \neq 0$ never enters — deterministic W kills term (I)
- CJIVE: zeros within-cluster blocks of J_t^{ξ} , eliminating spillovers within the cluster

First stage:

$$\mathbb{E}[p' C_{\text{JIVE}} p] = \text{tr}(C_{\text{JIVE}} \Omega), \quad \Omega_t = J_t^\xi \sigma_\xi^2 I (J_t^\xi)' + J_t^\eta \Sigma_{\eta,t} (J_t^\eta)'$$

Probability limit:

$$\text{plim}(\hat{\beta}^{\text{JIVE}} - \beta) = \frac{\sigma_\xi^2 \cdot \text{tr}(C_{\text{JIVE}} J^\xi)}{\text{tr}(C_{\text{JIVE}} \Omega)}$$

Validity: $\text{tr}(C_{\text{JIVE}} J^\xi) = 0$ — restricts both estimator (via C) **and** economic structure (via J^ξ).

Bartik: stochastic- W case

Labor supply: $\hat{\omega}_\ell = \tilde{\varphi} \hat{L}_\ell + \varepsilon_\ell$.

$$\hat{L}_{\ell n} = g_n^* + \frac{\sigma}{\sigma + \varphi} \varepsilon_\ell + \hat{\xi}_{\ell n} - \frac{\sigma}{\sigma + \varphi} \sum_{n'} s_{\ell n'} (g_{n'}^* + \hat{\xi}_{\ell n'})$$

Own-observation Jacobian:

$$[J_\varepsilon^{\text{own}}]_{n\ell} = \frac{\sigma}{\sigma + \varphi} \underbrace{\omega_{\ell n}}_{\text{location share of industry}}$$

Three BHJ assumptions zero out $J_\varepsilon^{\text{GE}}$:

1. **Immobile labor** — ε_ℓ does not redistribute across ℓ'
2. **No interregional trade** — supply shock in ℓ no effect on demand in ℓ'
3. **Residual is purely supply-side** — GE reallocation orthogonal to ε_ℓ

Let $H_\ell \equiv \sum_n s_{\ell n} \omega_{\ell n}$ (region ℓ 's exposure-weight self-overlap).

Trace.

$$\text{tr}(SJ_\varepsilon) = \sum_\ell \sum_n s_{\ell n} [J_\varepsilon]_{n\ell} = \frac{\sigma}{\sigma + \varphi} \sum_\ell H_\ell$$

Non-LOO bias (predetermined S , $\Sigma_\varepsilon = \sigma_\varepsilon^2 I$):

$$\mathbb{E} \left[\sum_\ell z_\ell \varepsilon_\ell \right] = \frac{\sigma \sigma_\varepsilon^2}{\sigma + \varphi} \sum_\ell \mathbb{E}[H_\ell]$$

LOO bias. Replace g_n with $g_{n,-\ell} \rightarrow$ diagonal of SJ_ε is zeroed:

$$\mathbb{E} \left[\sum_{\ell} z_{\ell}^{\text{LOO}} \varepsilon_{\ell} \right] = 0$$

Compare Hausman LOO bias $\lambda \sigma_{\xi}^2$: nonzero because λ creates off-diagonals that LOO can't reach. BHJ **assumes those away**.

(i) GE entries: $J_{\varepsilon}^{\text{GE}} \neq 0$. Mobile labor: $\varepsilon_{\ell} \rightarrow$ wage in $\ell \rightarrow$ migration to $\ell' \rightarrow \hat{L}_{\ell'n} \rightarrow$ contamination of g_n .

- Each entry $O(\frac{\varphi}{L})$, but row sum $O(\varphi)$ — does **not** shrink under large- L heuristic.

(ii) Off-diagonal Σ_{ε} . Group supply shocks: $\varepsilon_{\ell} = \sum_g \tilde{w}_{\ell g} \hat{v}_g + \hat{v}_{\ell}$.

- Couples $J_{\varepsilon}^{\text{own}}$ with off-diagonal Σ_{ε} — bias LOO doesn't address.

(iii) Endogenous shares. If $\text{Cov}(s_{\ell n}, \varepsilon_{\ell}) \neq 0$:

$$\bar{g}' c_{w\varepsilon} = \sum_n \bar{g}_n \sum_{\ell} \text{Cov}(s_{\ell n}, \varepsilon_{\ell}) \neq 0$$

Active **even when** $J_{\varepsilon} = 0$ — pure share-endogeneity bias.

Mobile labor adds a cross-region migration channel:

$$\hat{L}_{\ell n} = \text{BHJ terms} + \frac{\rho}{L} \sum_{\ell'} \varepsilon_{\ell'}$$

Jacobian gains a uniform off-diagonal term (analog of Hausman's $\frac{\lambda}{M}$):

$$\left[J_\varepsilon^{\text{AKM-1}} \right]_{n\ell} = \underbrace{\frac{\sigma}{\sigma + \varphi} \omega_{\ell n}}_{\text{BHJ diagonal}} + \underbrace{\frac{\rho}{L}}_{\text{migration spillover}}$$

Non-LOO bias: $\sigma_\varepsilon^2 \left[\frac{\sigma}{\sigma + \varphi} \sum_\ell \mathbb{E}[H_\ell] + \rho \right]$.

LOO bias. Diagonal zeroed, but $\frac{\rho}{L}$ survives — exactly as $\frac{\lambda}{M}$ does in Hausman:

$$\mathbb{E} \left[\sum_{\ell} z_{\ell}^{\text{LOO}} \varepsilon_{\ell} \right] = \sigma_{\varepsilon}^2 \sum_{\ell} \sum_n s_{\ell n} \left(\frac{\rho}{L} \right) (1 - \omega_{\ell n}) \rightarrow \rho \sigma_{\varepsilon}^2 \quad (L \rightarrow \infty)$$

ρ is the Bartik analog of λ

AKM-2: group residuals. $\varepsilon_\ell = \sum_g \tilde{w}_{\ell g} \hat{\nu}_g + \hat{\nu}_\ell \rightarrow$ off-diagonal Σ_ε .

- LOO zeros diagonal of SJ_ε , but off-diagonals couple to off-diagonal Σ_ε :

$$\mathbb{E} \left[\sum_\ell z_\ell^{\text{LOO}} \varepsilon_\ell \right] = \frac{\sigma}{\sigma + \varphi} \sum_g \sigma_{\nu_g}^2 \text{tr}(A_g B), \quad A_g = \tilde{w} \tilde{w}', B = SJ_\varepsilon^{\text{LOO}}$$

- CJIVE with cluster = group g removes it, but need to know groupings

AKM-3: endogenous shares. $\text{Cov}(s_{\ell n}, \varepsilon_{\ell}) \neq 0$ activates term (I):

$$\mathbb{E} \left[\sum_{\ell} z_{\ell} \varepsilon_{\ell} \right] = \underbrace{\sum_n \bar{g}_n \sum_{\ell} \text{Cov}(s_{\ell n}, \varepsilon_{\ell})}_{\text{term (I): survives LOO}} + \text{term (II)}$$

- LOO doesn't help; recentering / shock controls
- Bias scales with sector-specific mean growth $\bar{g}_n$

Appendix A11 in BHJ (JEP 2025): **“Can the instruments in Bartik (1991) and Card (2009) be valid without exogenous shares?** This depends on the underlying model, but probably not. We consider shift-share instruments with shifts constructed as national averages of endogenous local shifts correlated with the error terms, as in Bartik (1991) and Card (2009). The researcher might argue that these shifts proxy for some latent exogenous shifts. Here we show that the proxy error in the shifts is innocuous when the local shares are exogenous (and if there are many more observations than shifts), while otherwise the proxy error typically makes the instrument invalid. Thus, there is little value in focusing on the shifts for justifying the validity of Bartik (1991) and Card (2009) type instruments, except in a special case discussed below.”

Model	J_ε structure	Σ_ε	S	LOO bias
BHJ baseline	diag: $\frac{\sigma}{\sigma+\varphi}\omega_{\ell n}$	$\sigma_\varepsilon^2 I$	predetermined	0
AKM-1 (mobile labor)	1. uniform $\frac{\rho}{L}$	$\sigma_\varepsilon^2 I$	predetermined	$\rho\sigma_\varepsilon^2$
AKM-2 (group residuals)	diag (BHJ)	off-diag via groups	predetermined	$\frac{\sigma}{\sigma+\varphi} \sum_g \sigma_{\nu_g}^2 \text{tr}(A_g B)$
AKM-3 (endog shares)	diag (BHJ)	$\sigma_\varepsilon^2 I$	$\text{Cov}(s, \varepsilon) \neq 0$	$\sum_n \bar{g}_n \text{Cov}(s, \varepsilon)$
Hausman (GKL)	$\delta + \frac{\lambda}{M}$ diag, $\frac{\lambda}{M}$ off	$\sigma_\xi^2 I$	deterministic	$\lambda\sigma_\xi^2$

BHJ baseline = “ $\rho = 0$ ”. AKM-1’s ρ plays the role of Hausman’s λ .

Four channels, four corrections

Bias channel	Economic primitive	Correction
$J_{\varepsilon}^{\text{own}} \neq 0$ (own contamination)	$\delta \neq 0$ in pricing; $\sigma > 0$ in labor demand	LOO/JIVE; large- L heuristic; CJIVE within cluster
$J_{\varepsilon}^{\text{GE}} \neq 0$ (cross-unit GE)	λ uniform pricing; mobile labor; trade	CJIVE if spillovers within cluster; structural correction (calibrate σ)
Off-diag Σ_{ε} (correlated residuals)	coordinated demand shocks; group-specific supply shocks	split-sample g_n ; region/group FE; CJIVE with appropriate clusters
$\bar{g}' c_{w\varepsilon} \neq 0$ (share endogeneity)	cross-sector heterogeneity in $\bar{g}_n$ AND $\text{Cov}(s, \varepsilon) \neq 0$	recentering (Borusyak–Hull); share-control regressors; shares-as-IV (GPSS)

Hausman has **no analog** of recentering or share controls.

- JIVE weights are mechanically determined by the panel index
- $\text{Cov}(w_i, \varepsilon_i) = 0$ holds by **construction**
- Term (I) is structurally dead

The fact that $\bar{p} \neq 0$ in Hausman is informationally irrelevant.

In Bartik, three sets of corrections operate on three distinct primitives:

- location controls X_ℓ shrink $c_{w\varepsilon}$
- shock-level controls Q_n or recentering shrink $\bar{g}^\perp$
- LOO/CJIVE shrinks J_ε

Not substitutes when multiple channels bind.

Diagnostics

- Are estimates driven by one or two periods?
- Decompose the JIVE estimand into period-specific contributions:

$$\beta^{\text{JIVE}} = \sum_{t=1}^{T-1} w_t^{\text{JIVE}} b_t^{\text{JIVE}}$$

Per-period IV:

$$b_t^{\text{JIVE}} = \frac{\sum_{m \neq m'} \tilde{p}_{mt} \tilde{q}_{m't}}{\sum_{m \neq m'} \tilde{p}_{mt} \tilde{p}_{m't}}$$

Weights:

$$w_t^{\text{JIVE}} \propto M(M-1)(\bar{p}_t - \bar{p})^2 - M \widetilde{\text{Var}}_t(p)$$

1. **Plot weights w_t across periods.** Concentration signals fragile identification.
2. **Plot single-period b_t .** Under correct specification, all $b_t = \beta$. Dispersion signals heterogeneity or differential bias.
3. **Compare b_t^{2SLS} vs. b_t^{JIVE} .** Large differences signal strong within-period correlation.
4. **Leave-one-period-out sensitivity.**
5. **Flag negative JIVE weights.** Consider excluding or using CJIVE.

Bartik analog: Rotemberg weights identify high-leverage industries (GPSS).

Illustration: DellaVigna and Gentzkow (2019)

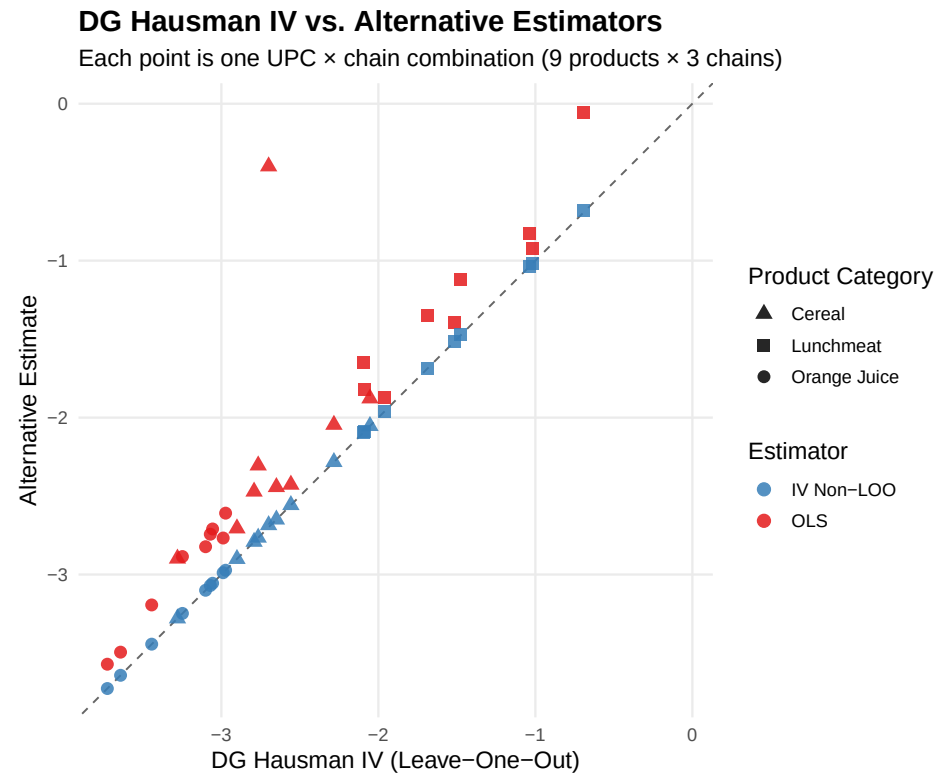
- Weekly store-level scanner data: 9 products (3 OJ, 3 cereals, 3 lunchmeats), 3 retail chains, 3,448 stores, 469 weeks (2006–2014)
- Demand equation per chain–product:

$$\log Q_{st} = \eta \log P_{st} + \alpha_{sy} + \gamma_{sw} + \varepsilon_{st}$$

- Instrument: chain–week mean of log prices
 - LOO mean (excludes focal store)
 - non-LOO mean (includes focal store)

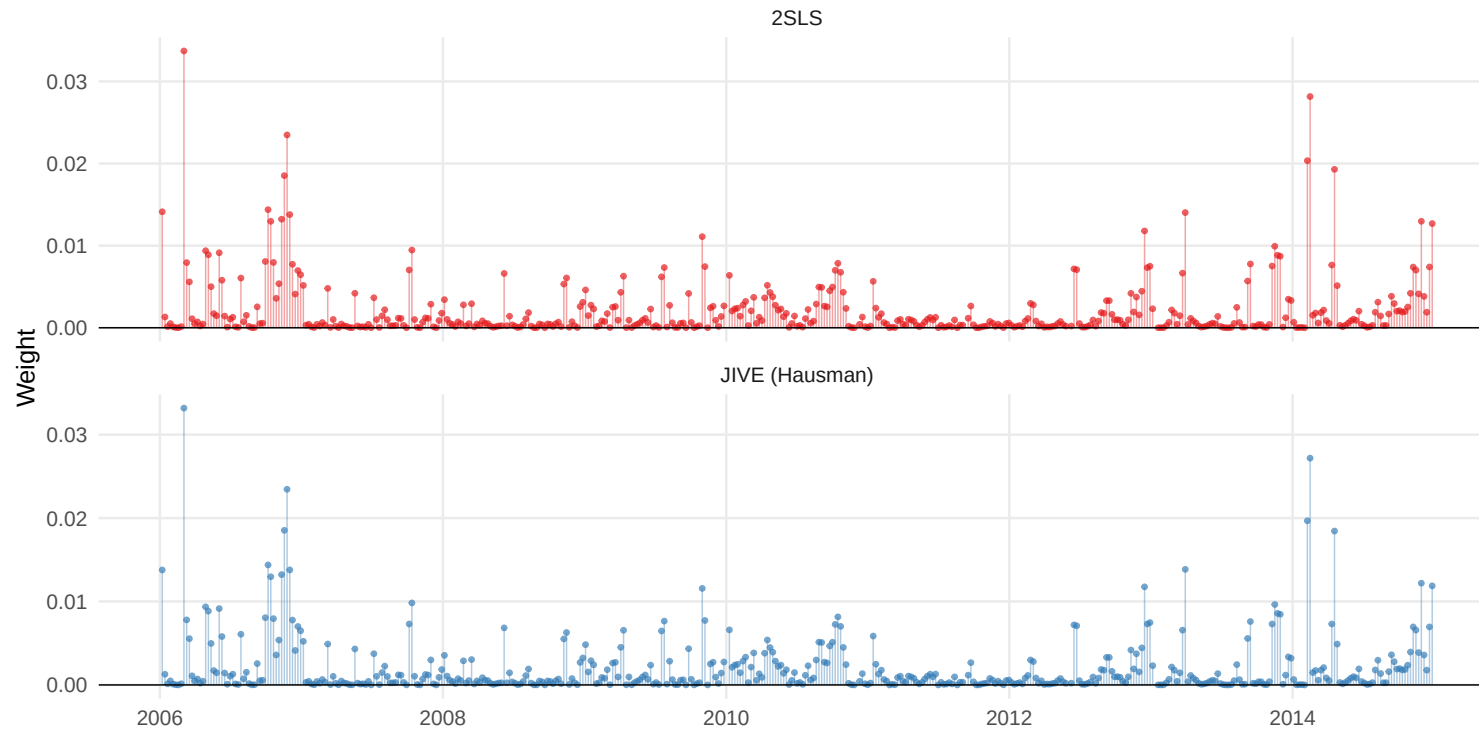
Chain	IV non-LOO	Week-dummy 2SLS	Diff	Balance
130_130	—3.0700	—3.0698	0.0002	93.4%
32_32	—2.9883	—2.9884	—0.0001	93.9%
79_79	—3.6419	—3.6415	0.0004	92.1%

Differences arise solely from missing store-week observations. UPC 2500005542.



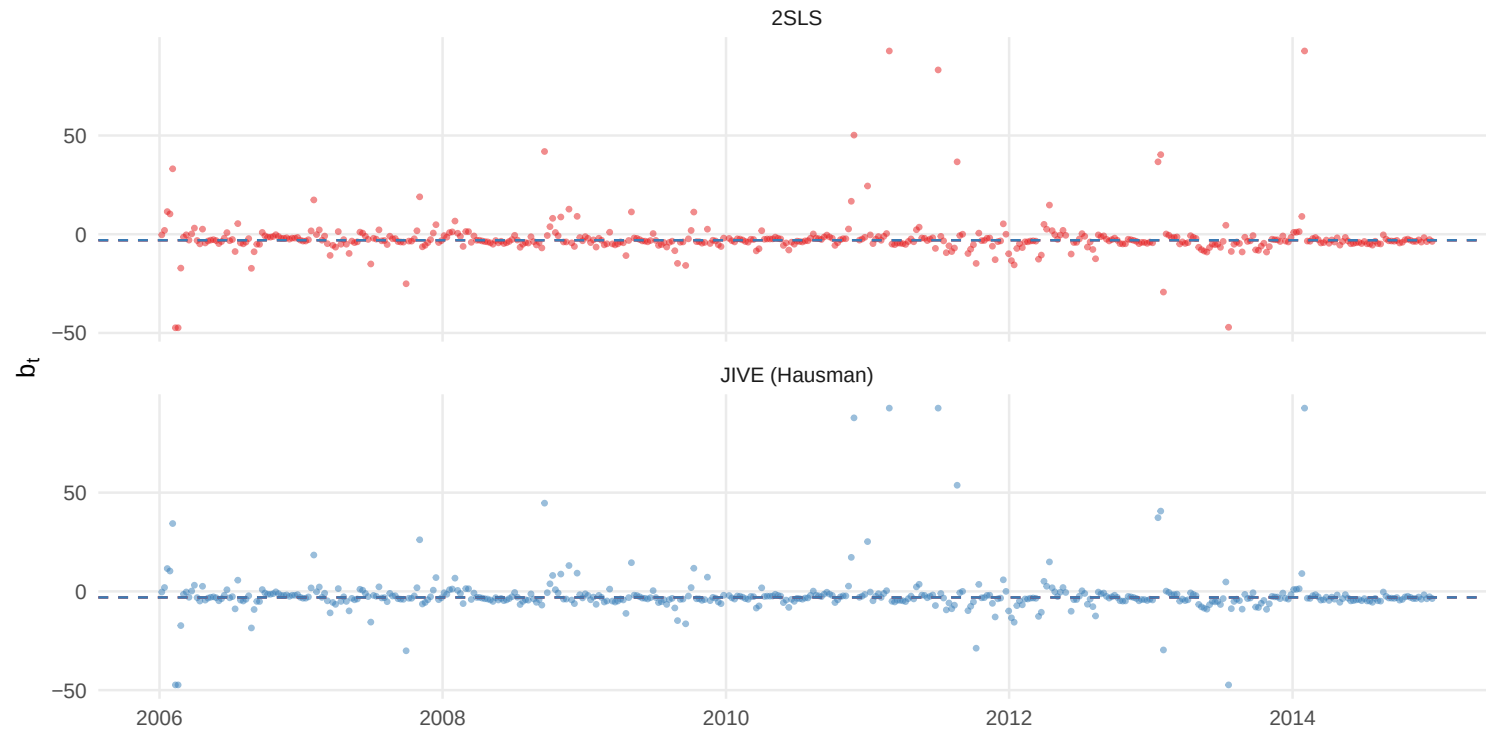
Diagnostic 1: Period Weights

UPC 2500005542, Chain 130_130 | 464 periods



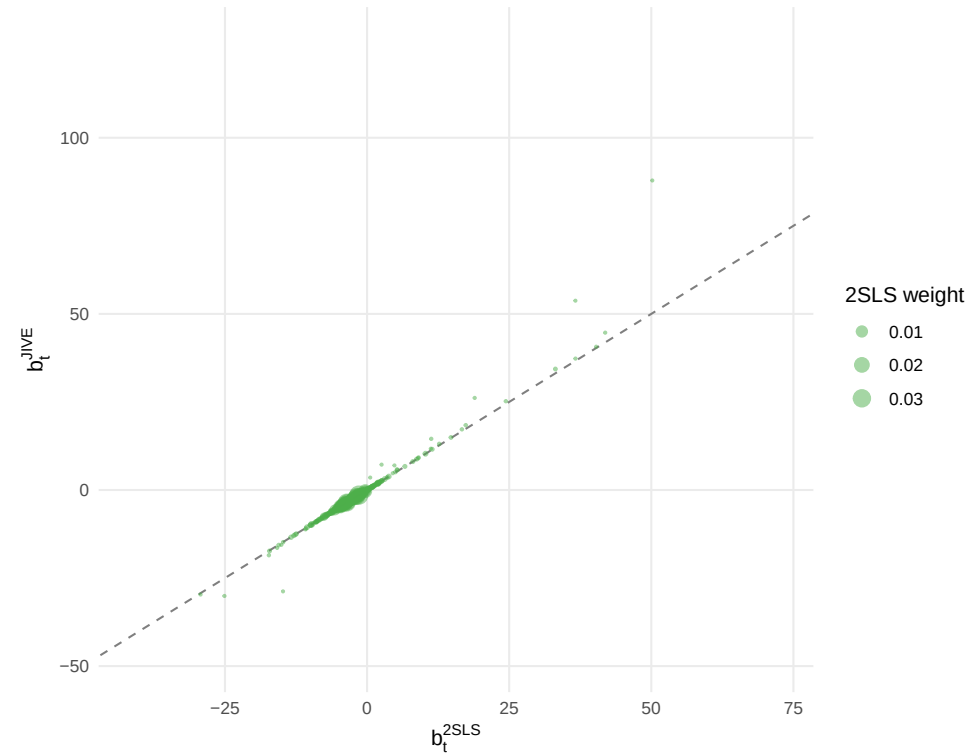
Diagnostic 2: Period-Specific Estimands

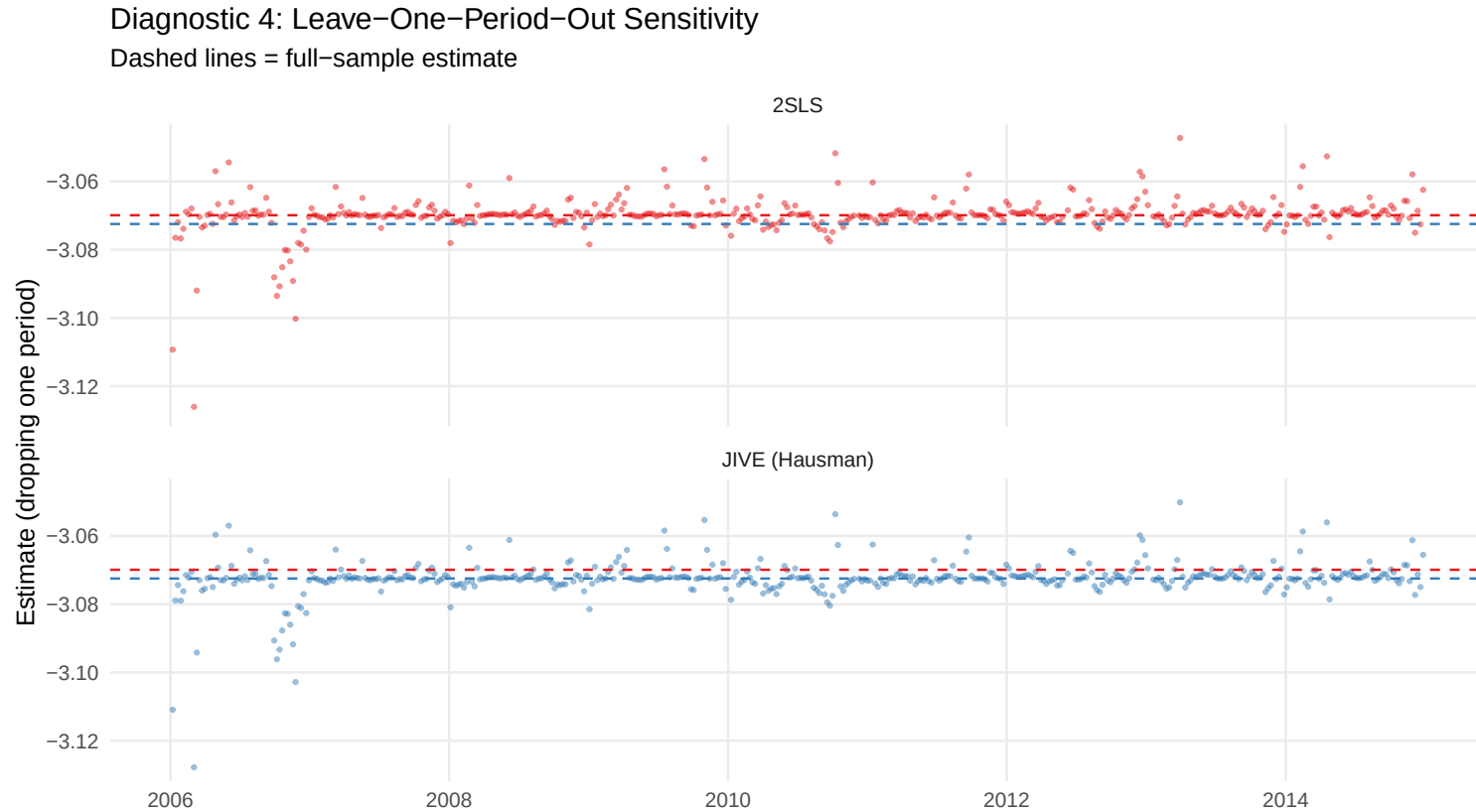
Dashed lines = overall estimate. Winsorized at 0.5th/99.5th pctile.



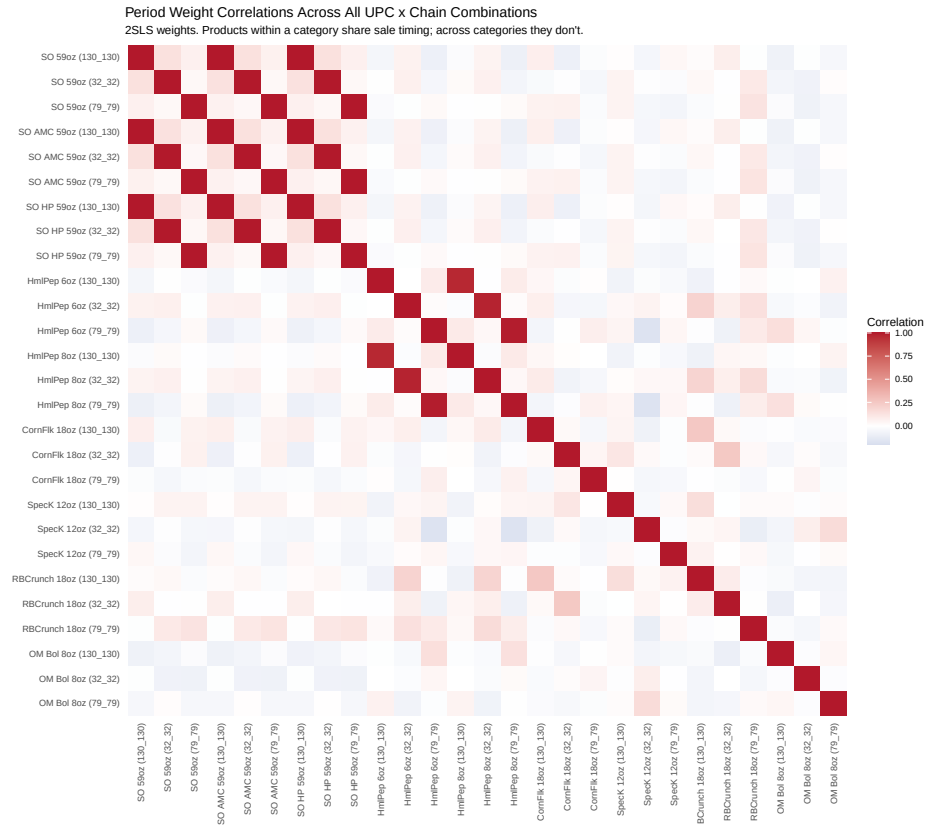
Diagnostic 3: 2SLS vs JIVE Period Estimands

Point size proportional to 2SLS weight. Axes trimmed at 0.5/99.5 pctile.





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[Bartik empirical illustration — TBD]

Open to suggestions!

- Hausman and shift-share are **empirical-proxy** designs: $z_i = w_i' g$, $g = h(\varepsilon, \eta)$ is an equilibrium outcome
- Unified bias formula: $\bar{g}' c_{w\varepsilon} + \text{tr}(\mathbb{E}[W J_\varepsilon] \Sigma_\varepsilon)$
 - Hausman: deterministic W kills term (I); only contamination matters
 - Bartik: both can bind; $\bar{g}_n$ heterogeneity drives share-endogeneity bias
- Four bias channels, four corrections — **not** substitutes when multiple bind
- Diagnostics: period decomposition (Hausman) / Rotemberg weights (Bartik) localize identifying variation